

EARLY WARNING SYSTEM FOR LANDSLIDES

A satellite-based network system that can monitor all the landslide-prone sites of the country from a central monitoring station and generate an early warning to respective places is now possible



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In India, landslide-prone regions are spread over Himalayas, Western Ghats and Nilgiris. These cover very large areas. The term 'landslide' is generally used to describe the down-slope of soil, rock and organic material under the influence of gravity. It causes damage to property, people and a variety of resources in the disaster areas. If the ongoing deformation of land/surface can be monitored and measured with less than one centimetre (or a few mm) accuracy, the prediction of a landslide is possible. The global positioning system (GPS)

technology is capable of monitoring sub-centimetre deformations of this ground movement. Many countries have experimented measurement of land deformations using the GPS technology. This technology has also been implemented to monitor landslide, deformation of dam structure and mine wall failure.

The main advantage of GPS sensors over conventional deformation monitoring sensors is that the GPS requires no line-of-sight between the stations. This enables the GPS to monitor the landslides even during unfavourable weather, either in real time or post-processing mode. Surface monitoring

using traditional surveying techniques is a relatively slow process that restricts the number of points that can be regularly monitored. As a supplement to existing geotechnical instrumentation, the GPS offers a reliable and efficient method for three-dimensional monitoring.

GPS technology has been successfully applied to a variety of deformation monitoring applications. This is due to its ease of use and very high accuracy when appropriate hardware, software and field procedures are implemented.

I, therefore, propose here a satellite-based network system that can